

American International University–Bangladesh (AIUB)

Department of Computer Science, Faculty of Science and Technology

Mid-Term Examination Suggestions

Faculty: Mejbah Ahammad
Semester: Summer 2025-2026

Email: ahammadmejbah@aiub.edu
Date: July 25, 2026

Answer all questions.

Total Marks: 30

(20 × 1.5 = 30)

1. A software company continuously updates its application to improve user experience by adding dark mode, voice search, and personalized recommendations without changing the software environment. Which maintenance type is being performed?

- A. Corrective Maintenance
- B. Adaptive Maintenance
- C. Perfective Maintenance
- D. Preventive Maintenance

Answer: C

2. Which characteristic best distinguishes software from hardware?

- A. Software wears out over time.
- B. Software is manufactured in factories.
- C. Software does not physically deteriorate.
- D. Software requires mechanical maintenance.

Answer: C

3. Which SDLC model is most suitable when software requirements are completely understood and very unlikely to change?

- A. Prototype
- B. Agile
- C. Waterfall
- D. DSDM

Answer: C

4. Which development model emphasizes building software module by module until the complete system is delivered?

- A. Incremental Model
- B. Spiral Model
- C. V-Model
- D. Prototype Model

Answer: A

5. Which requirement specifies that an online examination system must allow students to submit answers electronically?

- A. Functional Requirement
- B. Business Requirement
- C. Non-functional Requirement
- D. Technical Requirement

Answer: A

6. Which Requirements Engineering activity produces the official Software Requirements Specification (SRS)?

- A. Elicitation
- B. Specification
- C. Validation
- D. Maintenance

Answer: B

7. Which problem occurs when different stakeholders request conflicting software features?

- A. Ambiguous Requirements
- B. Requirement Conflict
- C. Verification Failure
- D. Software Crisis

Answer: B

8. Agile software development places the highest priority on—

- A. Extensive documentation
- B. Customer collaboration
- C. Fixed requirements
- D. Formal approvals

Answer: B

9. Which Agile practice helps teams quickly identify project delays through frequent customer feedback?

- A. Pair Programming
- B. Daily Scrum
- C. Iterative Development
- D. Code Inspection

Answer: C

10. Which XP value encourages developers to keep software design as simple as possible?

- A. Courage
- B. Simplicity
- C. Feedback
- D. Respect

Answer: B

11. Which XP role tracks project progress and identifies deviations from the development plan?

- A. Customer
- B. Coach
- C. Tracker
- D. Tester

Answer: C

12. Which Scrum ceremony demonstrates completed work to stakeholders at the end of each Sprint?

- A. Sprint Planning
- B. Sprint Review
- C. Sprint Retrospective
- D. Daily Scrum

Answer: B

13. During which Scrum event does the team decide which Product Backlog items will be implemented in the next Sprint?

- A. Sprint Planning
- B. Sprint Review
- C. Sprint Retrospective
- D. Product Refinement

Answer: A

14. In DSDM, which prioritization category represents requirements that are desirable but not essential?

- A. Must Have
- B. Should Have
- C. Could Have
- D. Won't Have

Answer: B

15. Which DSDM concept fixes project time and cost while allowing functionality to be adjusted?

- A. Continuous Integration
- B. Pair Programming
- C. Timeboxing
- D. Code Inspection

Answer: C

16. Which Feature Driven Development activity allocates software features to individual developers?

- A. Develop Overall Model
- B. Plan by Feature
- C. Build by Feature
- D. Design by Feature

Answer: B

17. Which FDD principle focuses on delivering small, client-valued features instead of large software modules?

- A. Feature-Centric Development
- B. Timeboxing
- C. Continuous Deployment
- D. Pair Programming

Answer: A

18. Which software myth incorrectly assumes that adding more developers always speeds up a delayed software project?

- A. Customer Myth

- B. Management Myth
- C. Developer Myth
- D. Maintenance Myth

Answer: B

19. Which software quality attribute ensures that software performs its intended functions correctly?

- A. Reliability
- B. Portability
- C. Maintainability
- D. Scalability

Answer: A

20. Which activity verifies that every software requirement has a corresponding test case before implementation begins?

- A. Shift-Left Testing
- B. Code Refactoring
- C. Continuous Integration
- D. Sprint Review

Answer: A